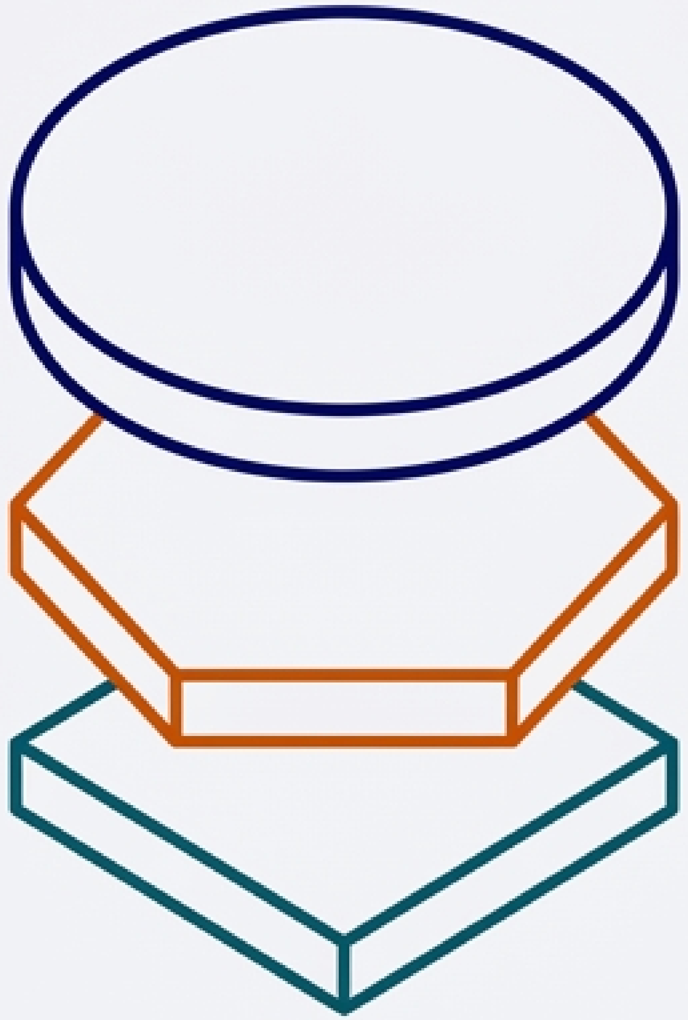


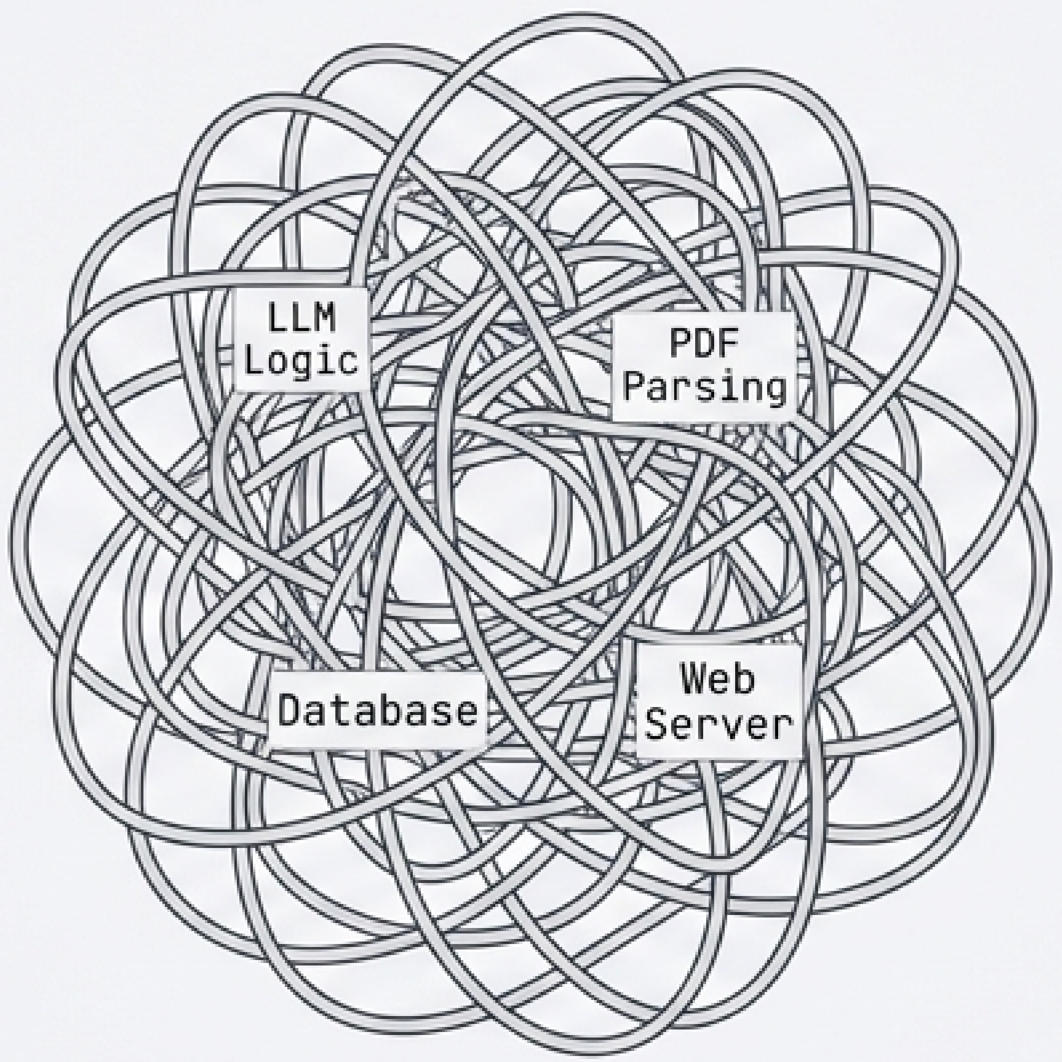
# AgentFlow Platform

## Scalable AI Architecture on Azure

DESIGN DOCUMENT: V1.0
TARGET: TECHNICAL ARCHITECTS & DECISION MAKERS
CONTENT: MICROSERVICES / DUAL INTERFACES / ASYNC PATTERNS

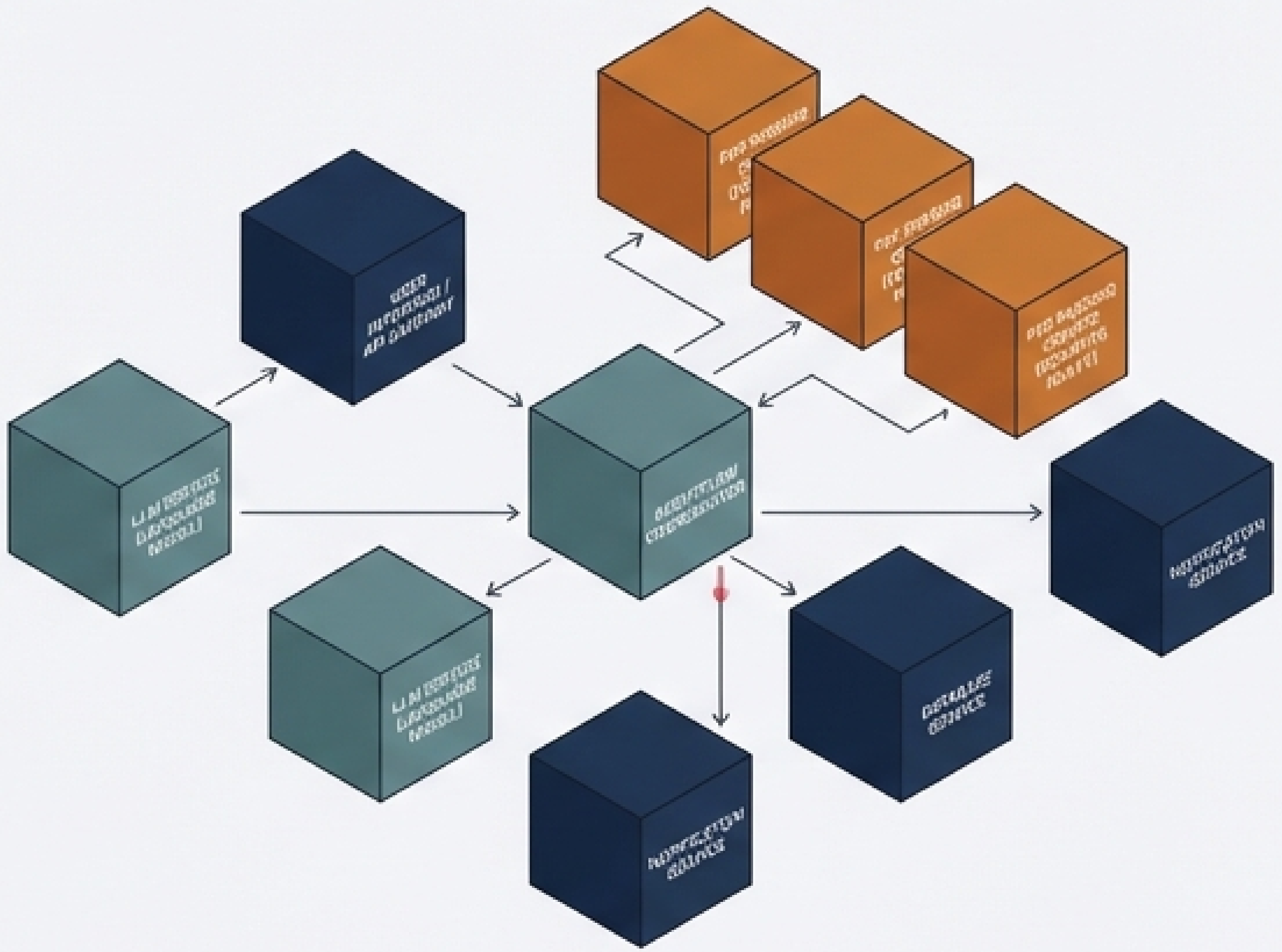
# The Shift: From Monolith to Microservices

## THE FRICTION



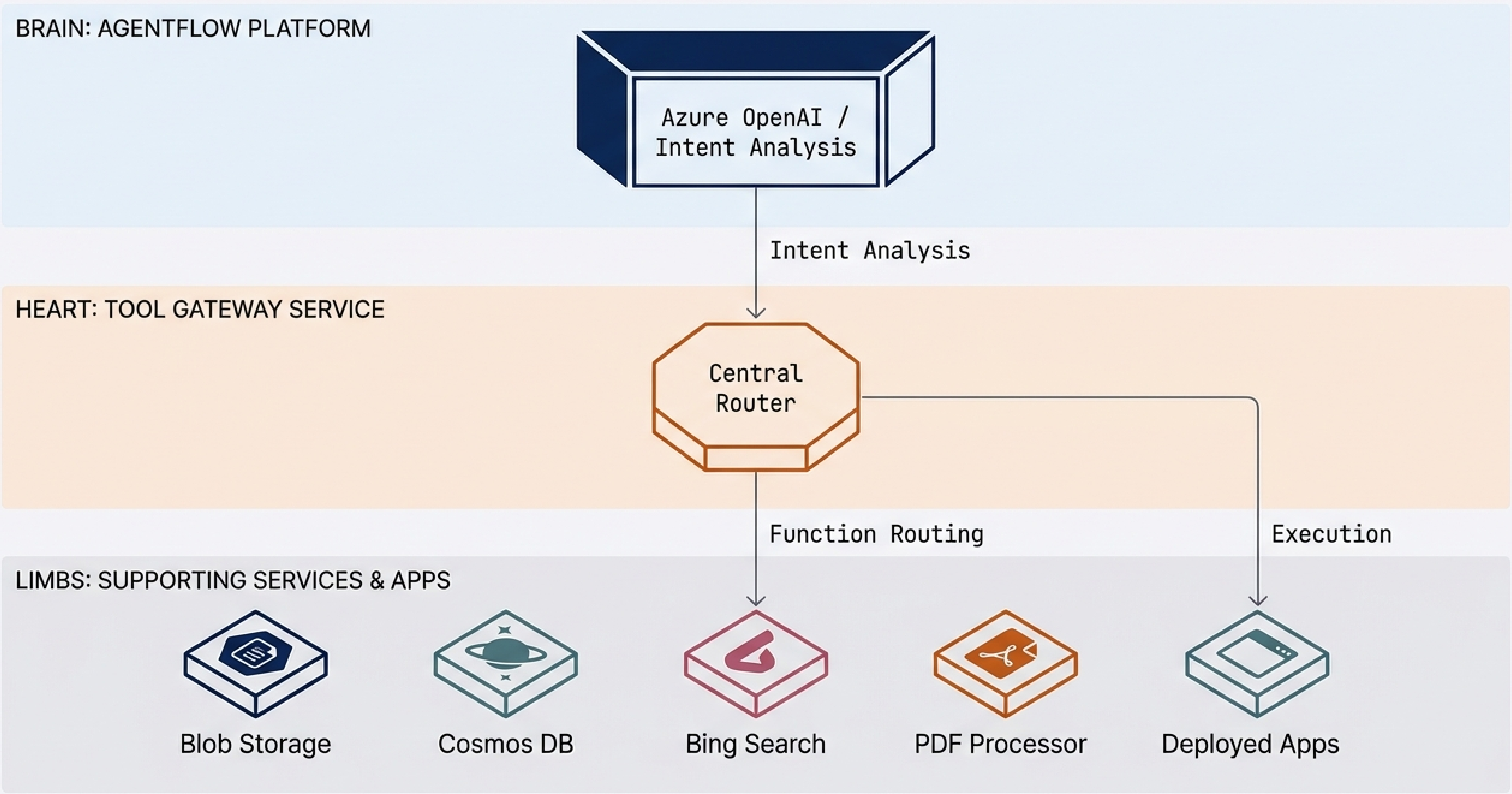

Tightly Coupled Monolith: Hard to Scale / Single Point of Failure.

## THE FIX

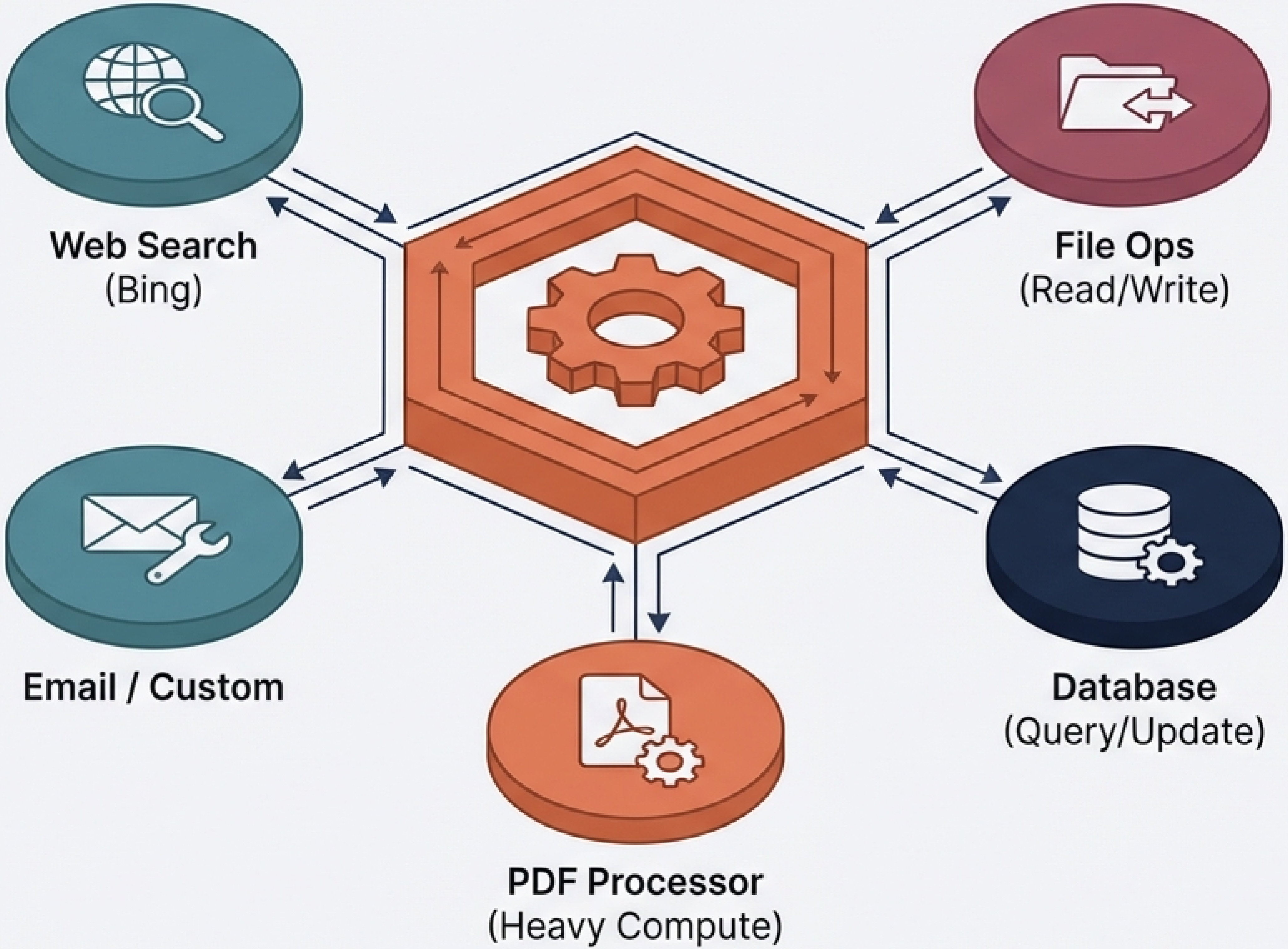


AgentFlow Architecture: Independent Scaling / Fault Isolation.

Decoupling heavy compute tasks (like PDF parsing) from user interactions prevents bottlenecks and allows components to evolve independently.


BRAIN: AGENTFLOW PLATFORM
Azure OpenAI / Intent Analysis
Intent Analysis
HEART: TOOL GATEWAY SERVICE
Central Router
Function Routing
Execution
LIMBS: SUPPORTING SERVICES & APPS
Blob Storage
Cosmos DB
Bing Search
PDF Processor
Deployed Apps
NotebookLM

# The Core Engine: Tool Gateway Service



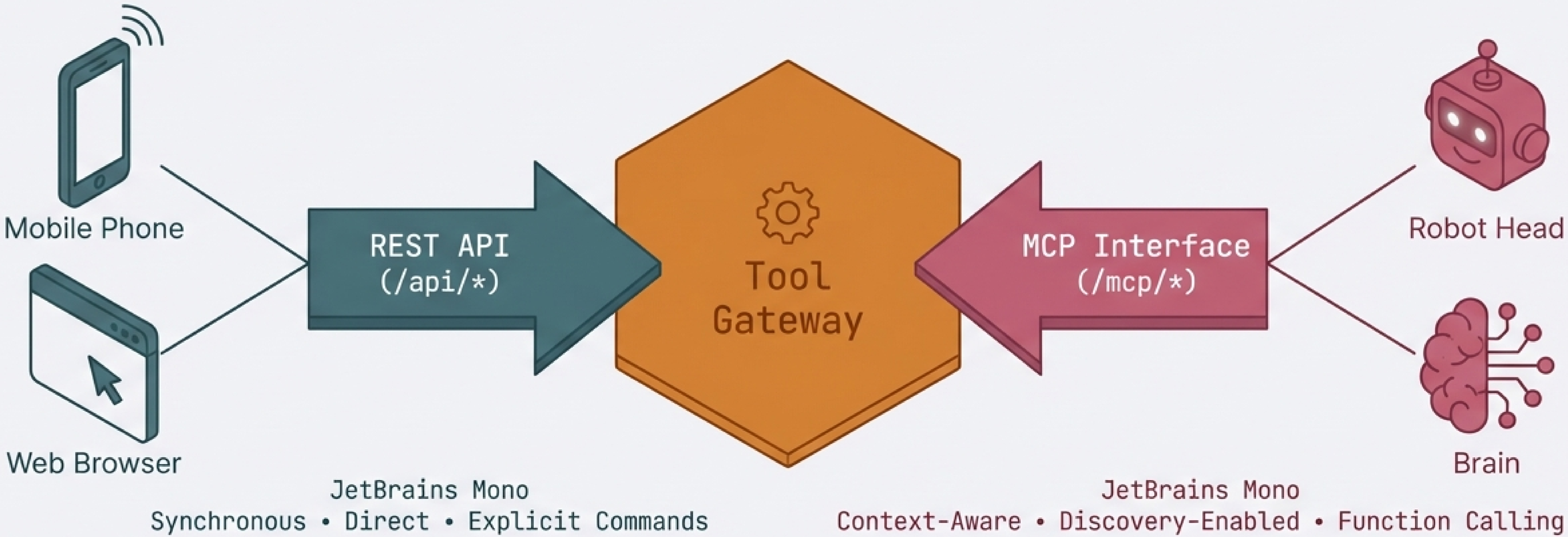

**ROLE:**

The central router that standardizes access to external capabilities.

**BENEFIT:**

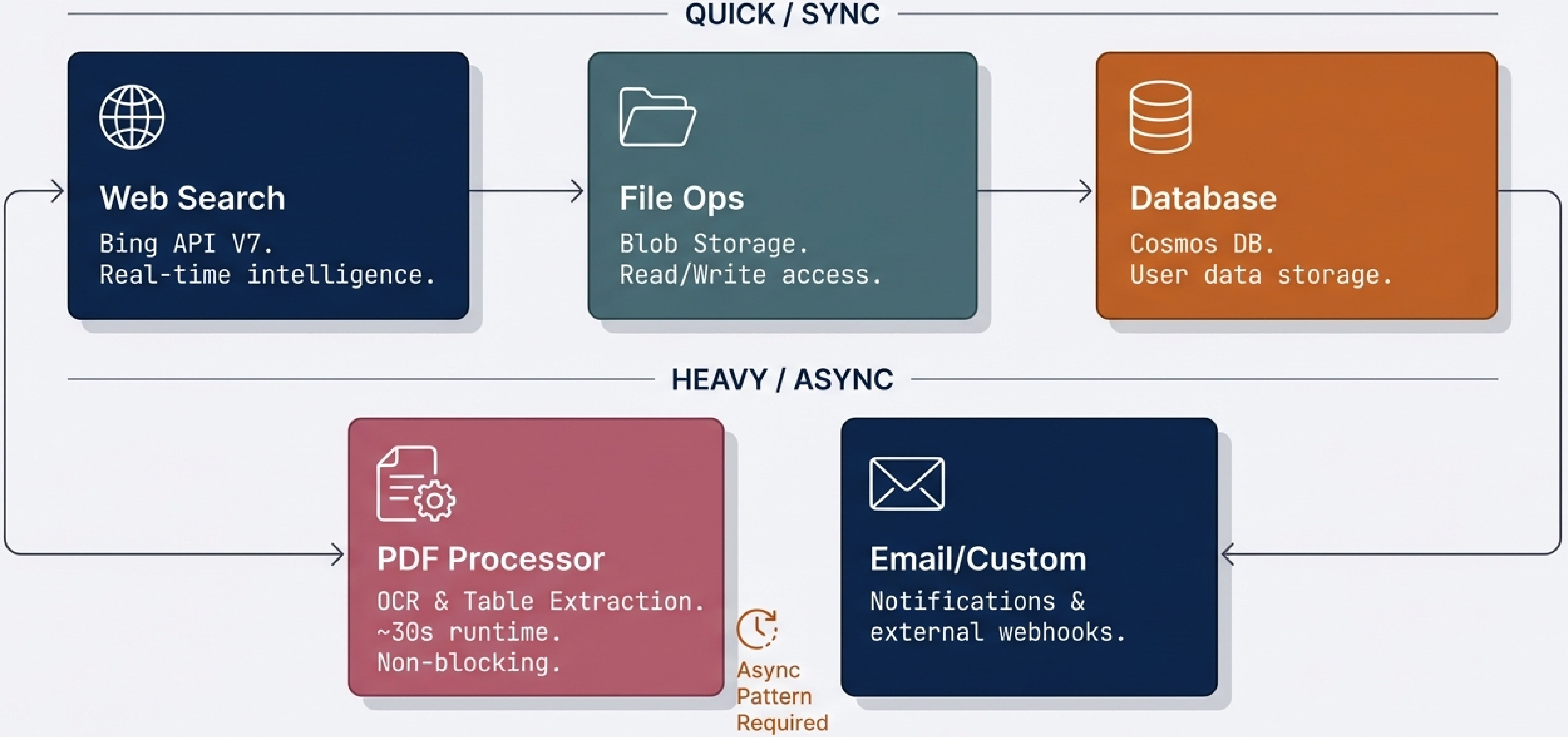
Removes tool management complexity from client applications. A **single integration point** for all "Smart" and "Dumb" tools.

# Dual Interfaces: Different Doors, Same House

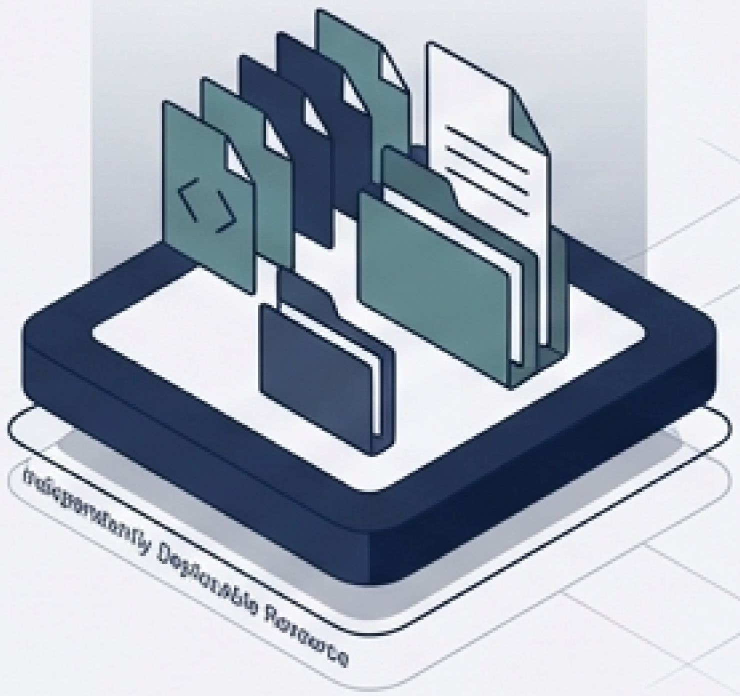



**Whether a React App** needs to run a command, or an AI Agent needs to discover tools, **both access the same underlying capabilities.**

# The Toolset: Synchronous vs. Asynchronous

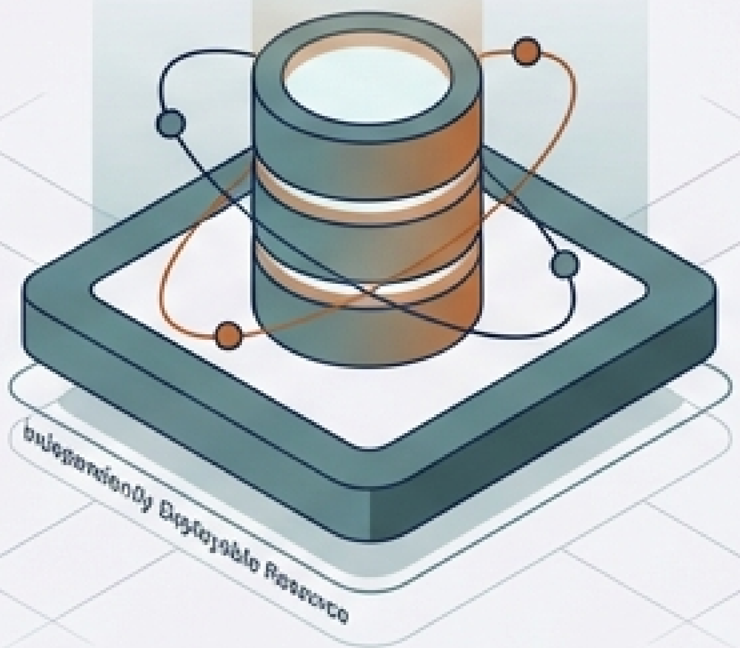


# Enterprise-Grade Supporting Infrastructure



**Blob Storage**

Unlimited Scale / Hot & Cold Tiers

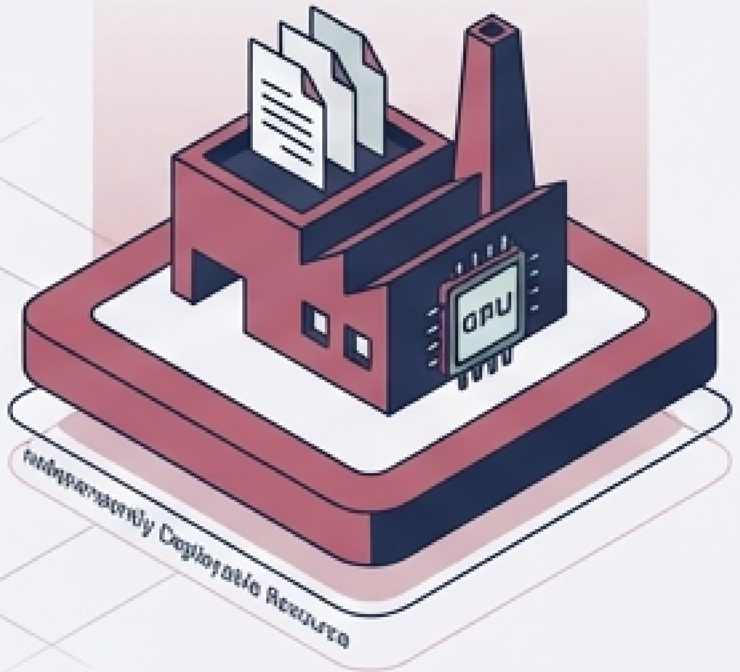


**Cosmos DB**

<10ms Latency / Multi-Region

**Bing Search API**

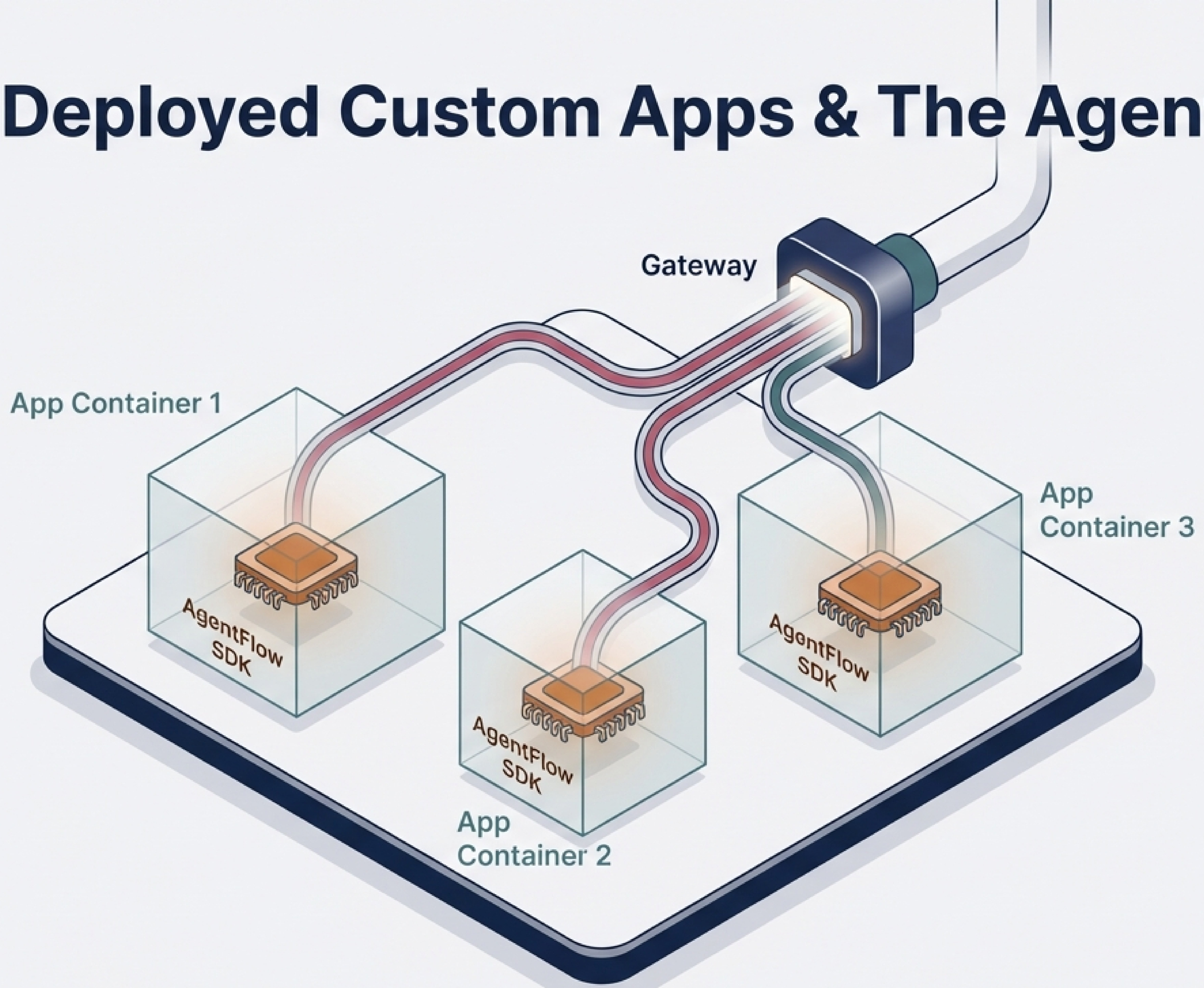
Real-time Web Intelligence




**PDF Processor Service**

Isolated Compute / OCR Service

# Deployed Custom Apps & The AgentFlow SDK

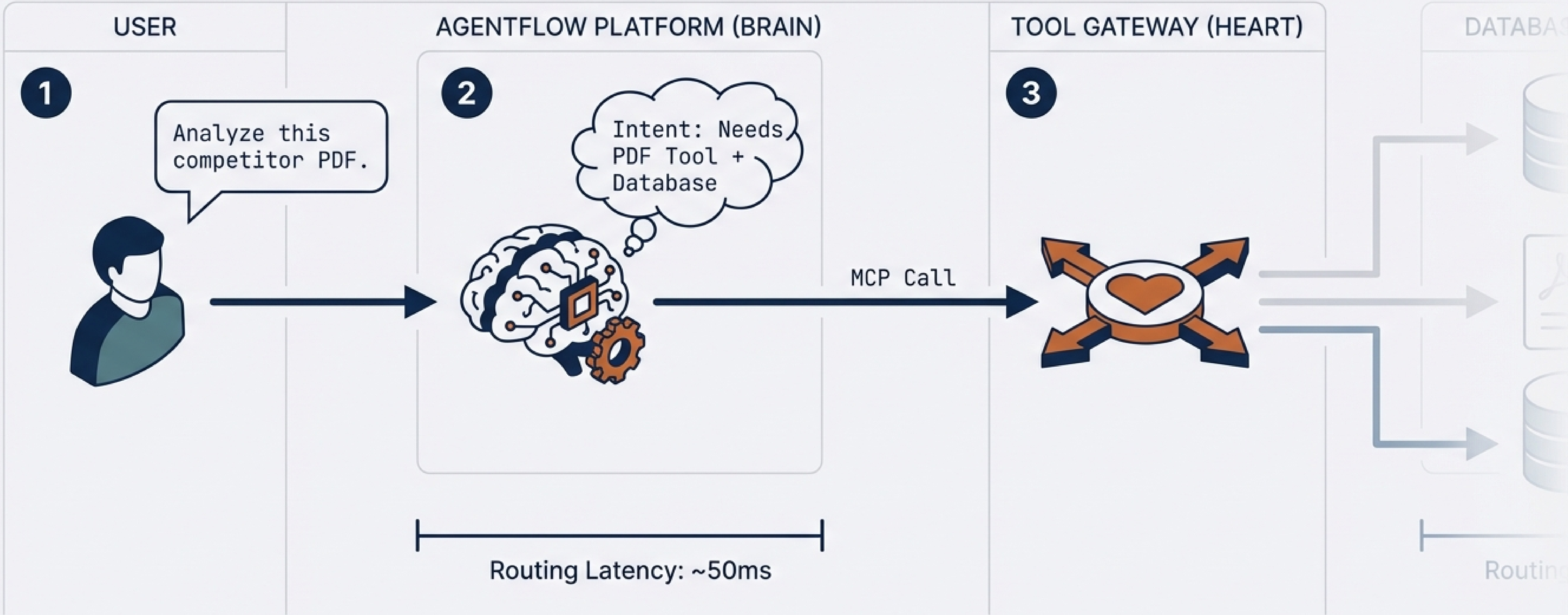



**FROM PROMPT TO CONTAINER:**

1. User describes app intent.
2. Platform generates Docker container.
3. App deploys to Azure Container Apps.
4. Included SDK handles auth and routing automatically.

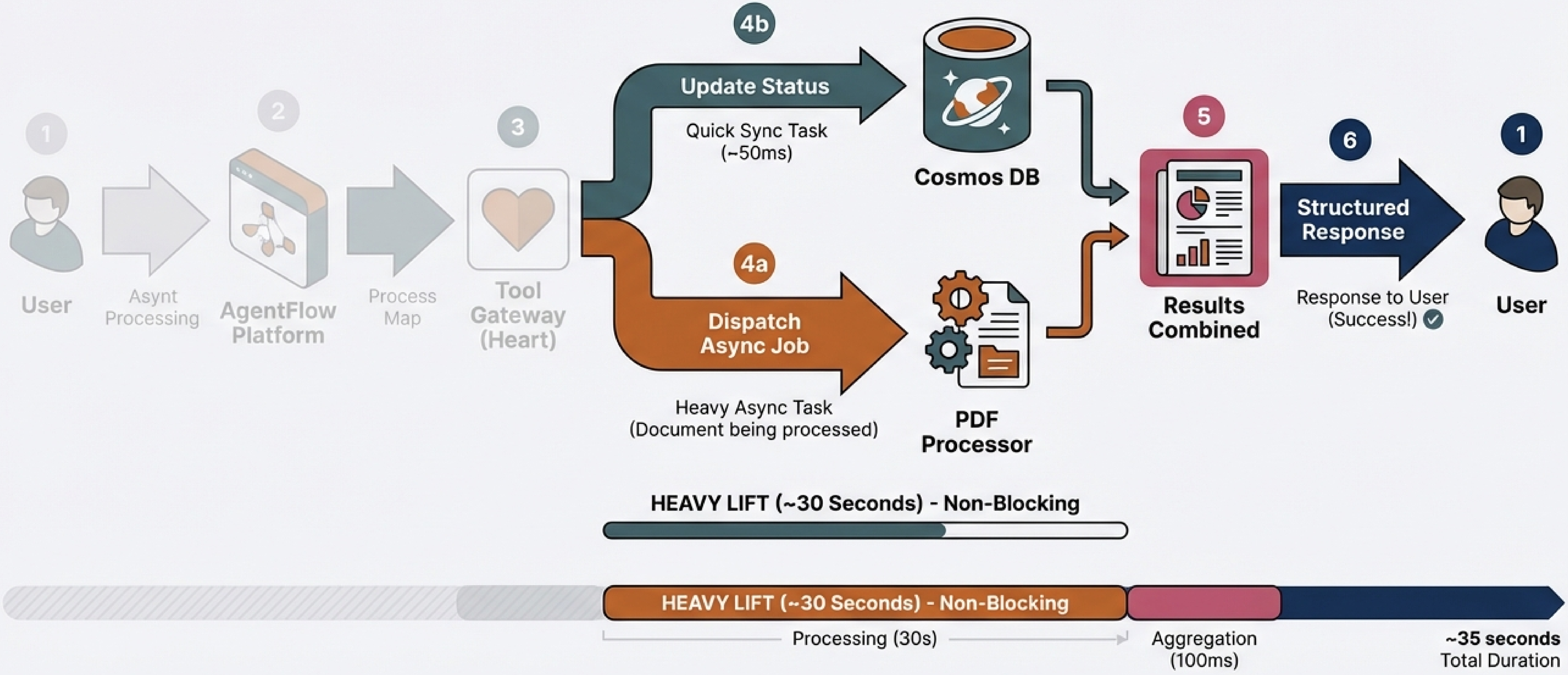
# Architecture in Action: The Workflow (Steps 1-3)

## Routing



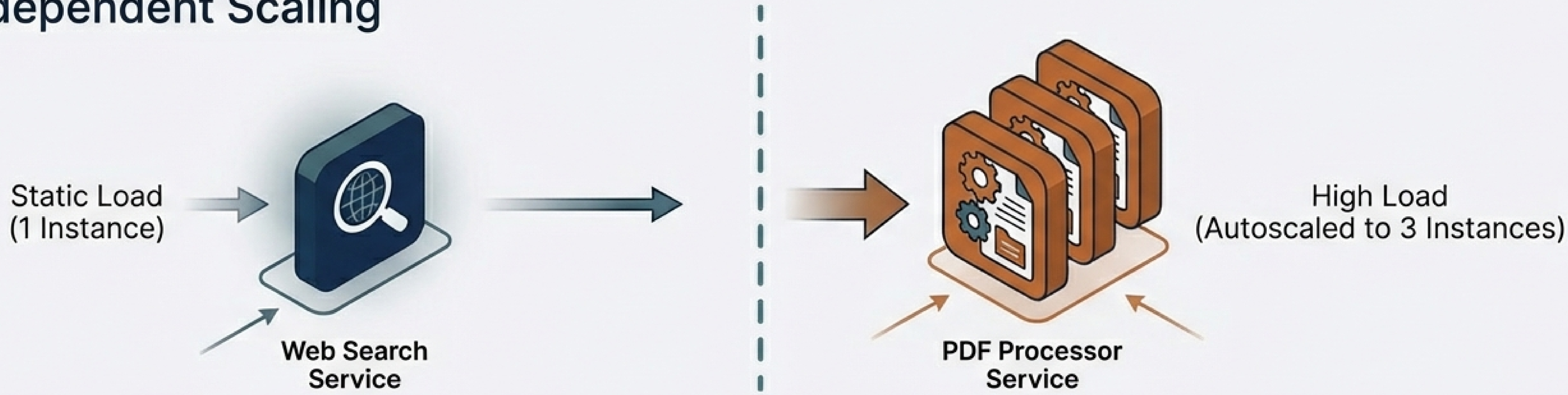

NotebookLM

# Architecture in Action: Async Processing (Steps 4-6)



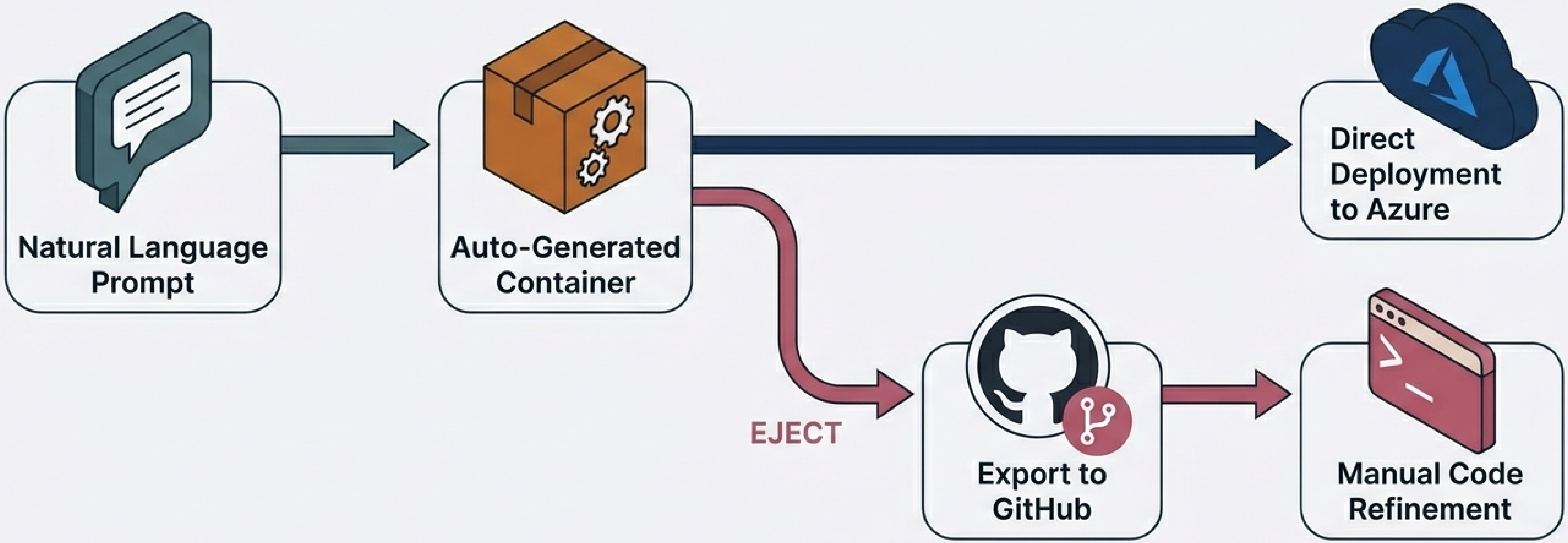
# Scalability & Resilience Patterns

## Independent Scaling



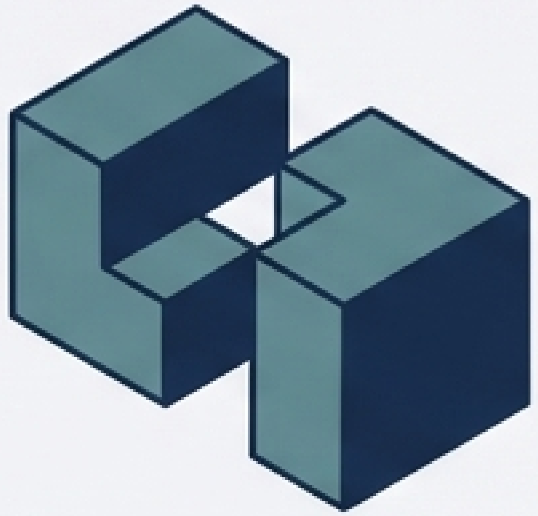

1. **INDEPENDENT SCALING:** Heavy compute services scale up without **over-provisioning the lightweight tools**.
2. **FAULT ISOLATION:** A crash in the PDF service does not take **down the Database** or **Search**.
3. **SCALE TO ZERO:** Services spin down completely when idle to save Azure costs.

# DevEx: From Prompt to Production



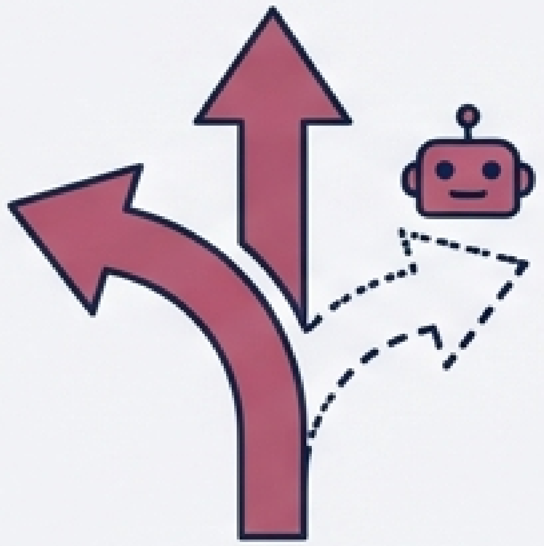

**Not a black box.** Export generated code to GitHub for manual hardening, security auditing, and productionalization.

# Strategic Architectural Advantages



## MODERN MICROSERVICES

Decoupled architecture replaces monolithic spaghetti code.

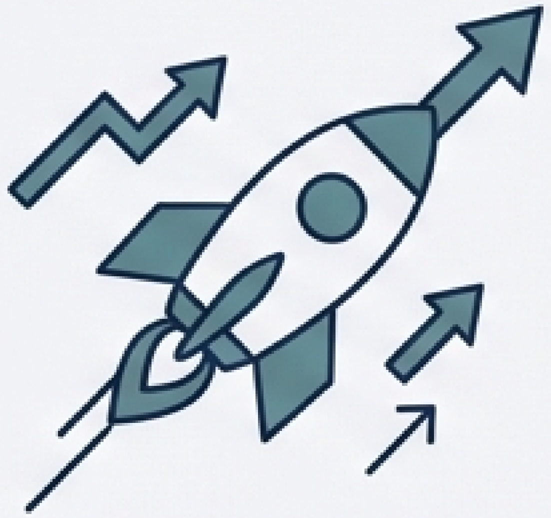


## HYBRID INTERFACES

REST for apps, MCP for AI. Future-proofed for new agents.

## ENTERPRISE FOUNDATION

Built on Azure PaaS (Container Apps, Cosmos, Blob) for reliability.



## RAPID DEPLOYMENT

Prompt-to-App generation with zero-downtime updates.

# Build Fast. Scale Smart.

## Deploy Anywhere.

AgentFlow combines the power of Azure infrastructure, microservices agility, and native AI discovery into a single, scalable platform.

AgentFlow + Azure

# Appendix: Technical Specifications

| CORE COMPONENTS | INTERFACE STANDARDS |
|---|---|
| **Orchestrator:** Azure Container Apps (KEDA-enabled) | **Gateway API:** REST (OpenAPI 3.0) |
| **Database:** Azure Cosmos DB (NoSQL API) | **AI Protocol:** Model Context Protocol (MCP 1.0) |
| **Storage:** Azure Blob Storage (Standard_LRS) | **Search Provider:** Bing Web Search API v7 |
| **AI Model:** Azure OpenAI Service (GPT-4o) | **Async Eventing:** Azure Event Grid |

Architecture valid as of System Design v1.0